

Numerical Results for meeting 2026-03-20

ATM Put BGK Path Comparison

ATM path-count experiment with Euler steps fixed at 100 and training paths varied from 50 to 100,000.

Parameters

- $S_0 = 100$
- $K = 100$
- $r = 0.0$
- $T = 1.0$
- $\theta = 0.078$
- $\kappa_1 = 5.5$
- $\kappa_2 = 0.1$
- $v_0 = 0.114$
- $v_{p0} = 0.110$
- $\delta_1 = 0.94$
- $\delta_2 = 0.94$
- $\xi_1 = 2.689$
- $\xi_2 = 0.502$
- $\rho_{12} = -0.982$
- $\rho_{13} = -0.727$
- $\rho_{23} = 0.59$
- Exercise dates = 100
- Euler steps = 100
- Path values = 50, 250, 500, 1,000, 5,000, 10,000, 20,000, 50,000, 80,000, 100,000
- Benchmark seeds = 2026, 2103
- Hybrid seeds = 2026, 2103

Reference Prices

Reference	Direct	Low
Benchmark reference	11.328264	11.309356

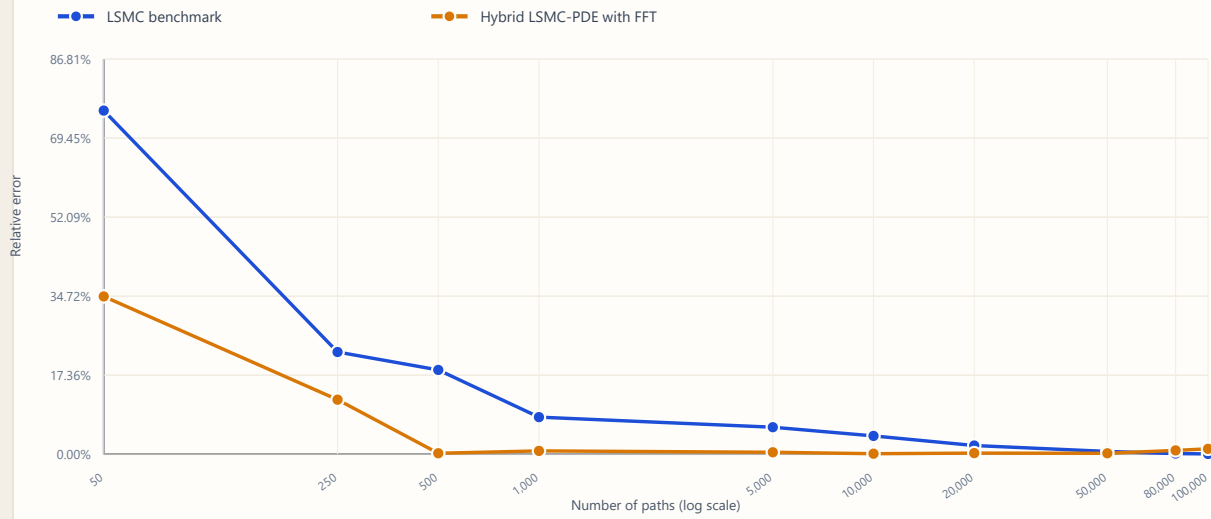
Direct Relative Error

ATM put: Direct relative error

Reference taken from bgk_gdmr_comparison.md; varying-path experiment curves use Euler 100 in the unchanged Final Code scripts.

BGK setup with $S_0 = 100$, $K = 100$, exercise dates = 100, path axis shown on log scale.

Relative errors are measured against the fixed ATM put benchmark direct reference value from the BGK note. Point estimates only.



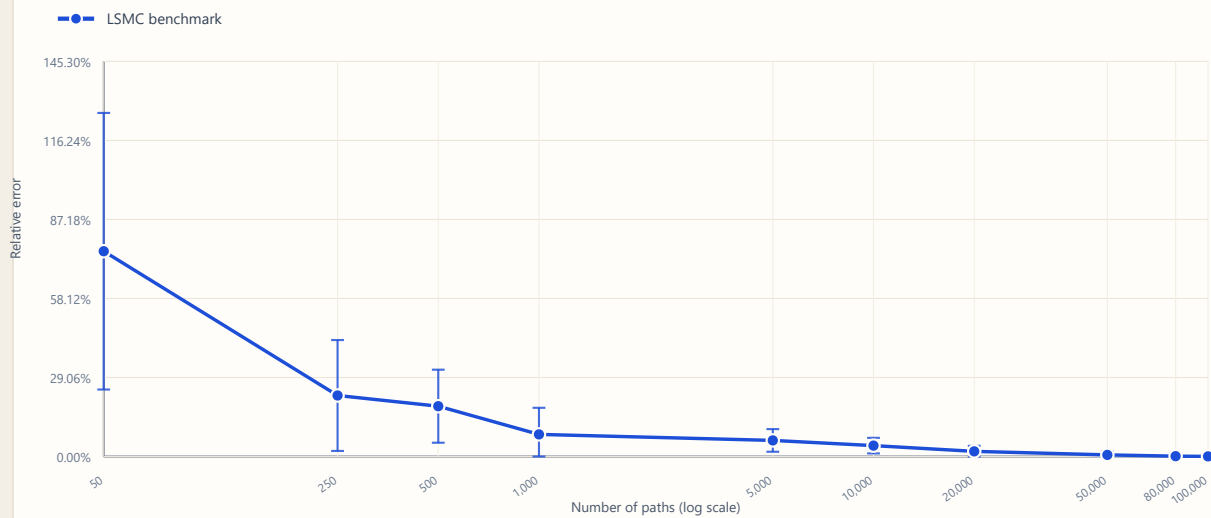
LSMC Direct Relative Error with 95% CI

ATM put: LSMC benchmark direct relative error

Reference taken from bgk_gdmr_comparison.md; only the LSMC benchmark direct curve is shown here.

BGK ATM setup with $S_0 = 100$, $K = 100$, exercise dates = 100, Euler steps = 100.

Relative errors are measured against the fixed ATM put benchmark direct reference. Pointwise 95% CI bars shown.



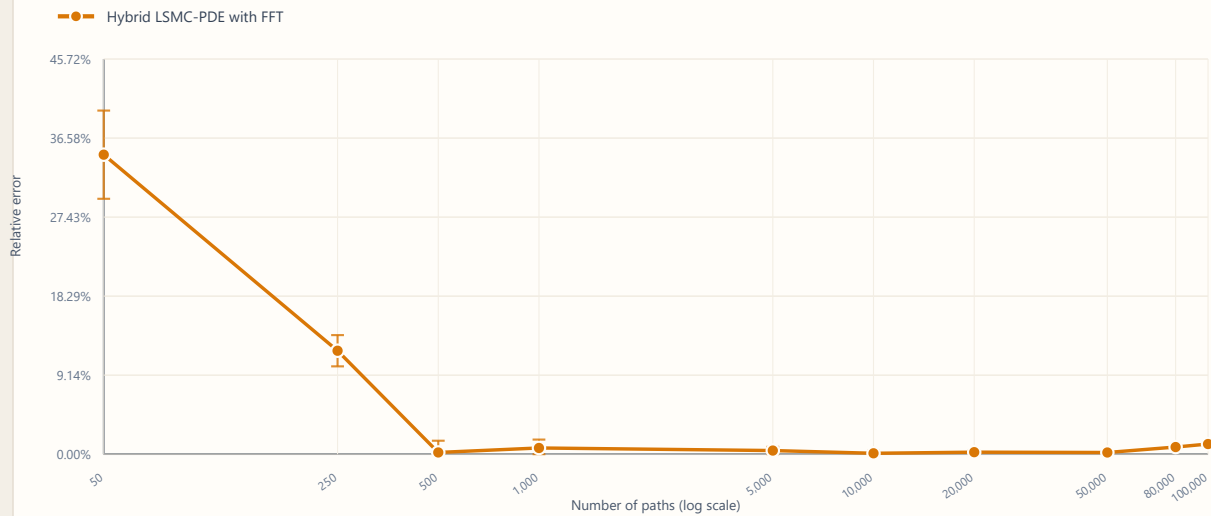
Hybrid Direct Relative Error with 95% CI

ATM put: Hybrid LSMC-PDE with FFT direct relative error

Reference taken from bgk_gdmr_comparison.md; only the Hybrid LSMC-PDE with FFT direct curve is shown here.

BGK ATM setup with $S_0 = 100$, $K = 100$, exercise dates = 100, Euler steps = 100.

Relative errors are measured against the fixed ATM put benchmark direct reference. Pointwise 95% CI bars shown.



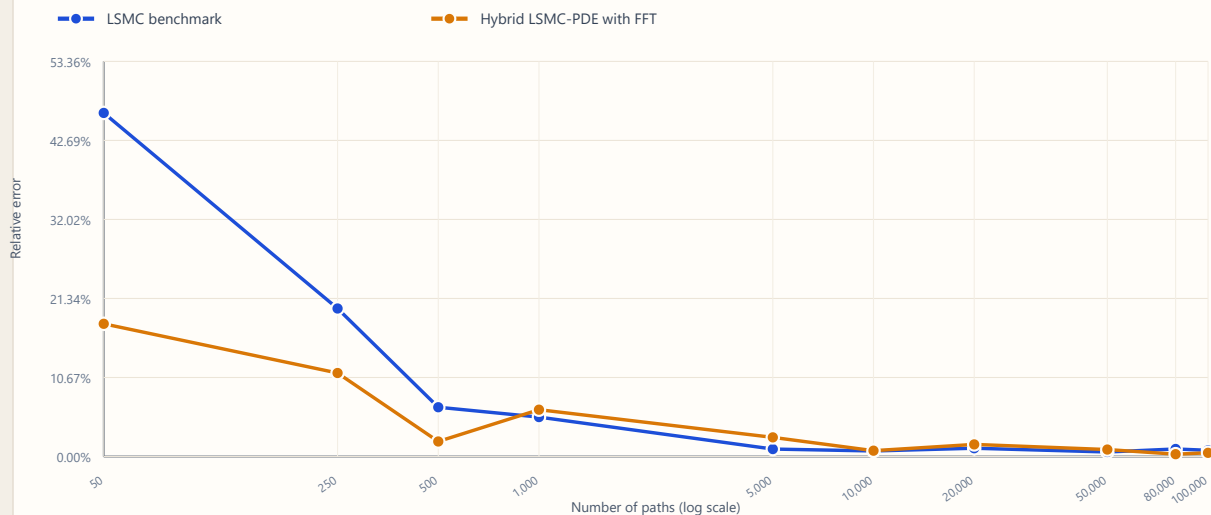
Low Relative Error

ATM put: Low relative error

Reference taken from bgk_gdmr_comparison.md; varying-path experiment curves use Euler 100 in the unchanged Final Code scripts.

BGK setup with $S_0 = 100$, $K = 100$, exercise dates = 100, path axis shown on log scale.

Relative errors are measured against the fixed ATM put benchmark low reference value from the BGK note. Point estimates only.



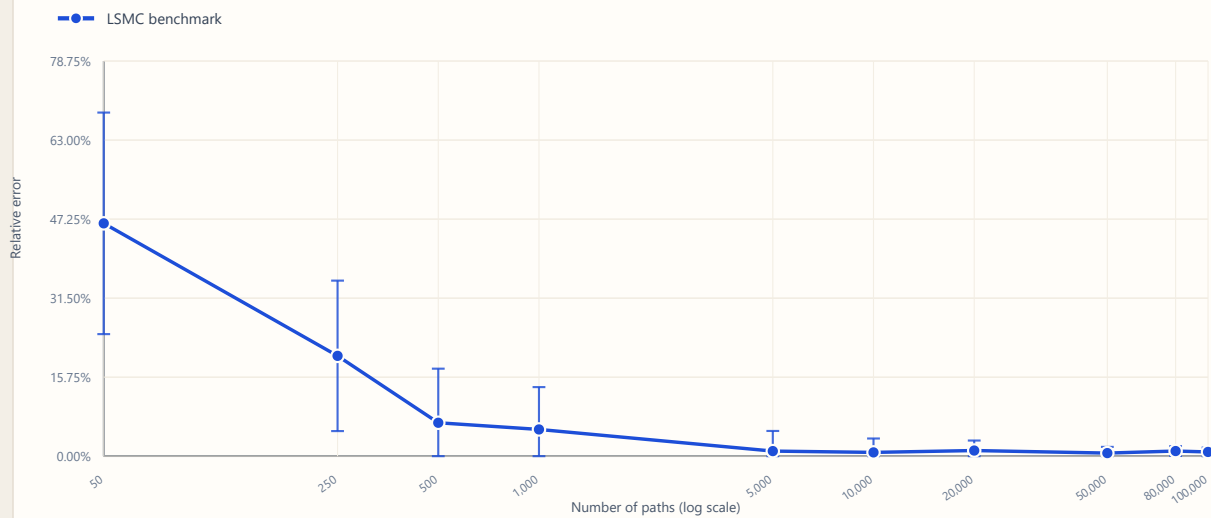
LSMC Low Relative Error with 95% CI

ATM put: LSMC benchmark low relative error

Reference taken from bgk_gdmr_comparison.md; only the LSMC benchmark low curve is shown here.

BGK ATM setup with $S_0 = 100$, $K = 100$, exercise dates = 100, Euler steps = 100.

Relative errors are measured against the fixed ATM put benchmark low reference. Pointwise 95% CI bars shown.



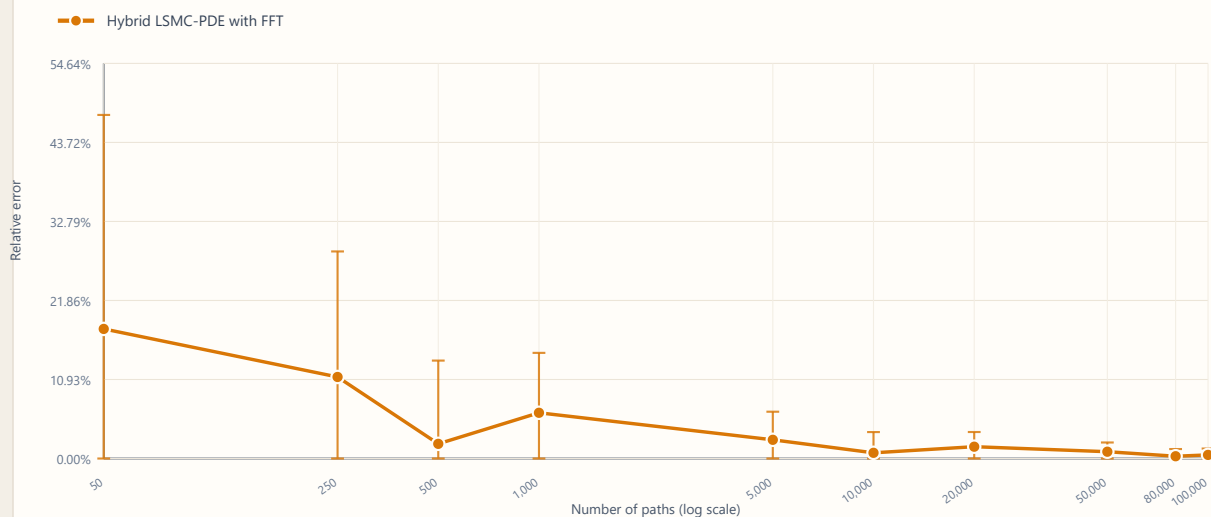
Hybrid Low Relative Error with 95% CI

ATM put: Hybrid LSMC-PDE with FFT low relative error

Reference taken from bgk_gdmr_comparison.md; only the Hybrid LSMC-PDE with FFT low curve is shown here.

BGK ATM setup with $S_0 = 100$, $K = 100$, exercise dates = 100, Euler steps = 100.

Relative errors are measured against the fixed ATM put benchmark low reference. Pointwise 95% CI bars shown.



Final gDMR Package Comparison

Single benchmark-versus-hybrid comparison under the final parameter set with $r = 0.03$, $T = 1.0$, and $\text{delta1} = \text{delta2} = 0.50$.

Parameters

- $S_0 = 100$
- $K = 100$
- $r = 0.03$
- $T = 1.0$
- $\theta = 0.04$
- $\kappa_1 = 2.0$
- $\kappa_2 = 1.0$
- $v_0 = 0.04$
- $v_{p0} = 0.04$
- $\delta_1 = 0.50$
- $\delta_2 = 0.50$
- $\xi_1 = 0.35$
- $\xi_2 = 0.20$
- $\rho_{12} = 0.20$
- $\rho_{13} = 0.10$
- $\rho_{23} = 0.10$
- Exercise dates = 100
- Euler steps = 600
- Benchmark paths = 1,000,000
- Benchmark low paths = 1,000,000
- Hybrid paths = 20,000
- Hybrid low paths = 20,000
- Hybrid asset points = 181
- Hybrid asset low factor = 0.35
- Hybrid asset high factor = 3.00
- Hybrid volatility quantile = 0.997
- Hybrid FST pad factor = 4
- Hybrid FST batch size = 256
- Benchmark seeds = 2026, 2103
- Hybrid seeds = 2026, 2103

Prices

Method	Direct	Low
Benchmark LSMC	6.294204	6.309298
Hybrid LSMC-PDE with FFT	6.300429	6.271978

Metrics

Metric	Direct	Low
Benchmark SE	0.008024	0.008035
Hybrid SE	0.000976	0.006975
Benchmark relative error	0.000%	0.000%
Hybrid relative error	0.099%	0.592%

Direct-Low Gap

Benchmark LSMC	Hybrid LSMC-PDE with FFT
+0.240%	-0.452%

BGK r=0.03, T=1 Comparison

BGK comparison across ATM, ITM, and OTM with `r = 0.03` and `T = 1.0`.

Parameters

- `S0 = 100`
- `r = 0.03`
- `T = 1.0`
- `theta = 0.078`
- `kappa1 = 5.5`
- `kappa2 = 0.1`
- `v0 = 0.114`
- `vp0 = 0.110`
- `delta1 = 0.94`
- `delta2 = 0.94`
- `xi1 = 2.689`
- `xi2 = 0.502`
- `rho12 = -0.982`
- `rho13 = -0.727`
- `rho23 = 0.59`
- `Scenarios = ATM (K = 100), ITM put (K = 110), OTM put (K = 90)`
- `Exercise dates = 100`
- `Euler steps = 600`
- `Benchmark paths = 1,000,000`
- `Benchmark low paths = 1,000,000`
- `Hybrid paths = 20,000`
- `Hybrid low paths = 20,000`
- `Hybrid asset points = 181`
- `Hybrid asset low factor = 0.35`
- `Hybrid asset high factor = 3.00`
- `Hybrid volatility quantile = 0.997`
- `Hybrid FST pad factor = 4`
- `Hybrid FST batch size = 256`
- `Benchmark seeds = 2026, 2103`
- `Hybrid seeds = 2026, 2103`

Scenario Summary

Direct

Scenario	K	Benchmark Direct	Hybrid Direct	Hybrid Direct Relative Error
ATM	100	10.373824	10.286746	0.839%
ITM Put	110	15.367020	15.311325	0.362%
OTM Put	90	6.708403	6.690825	0.262%

Low

Scenario	K	Benchmark Low	Hybrid Low	Hybrid Low Relative Error
ATM	100	10.354497	10.376803	0.215%
ITM Put	110	15.356918	15.437225	0.523%
OTM Put	90	6.690326	6.757117	0.998%

Metrics

Scenario	Benchmark Direct SE	Benchmark Low SE	Hybrid Direct SE	Hybrid Low SE
ATM	0.014582	0.014565	0.011403	0.100545
ITM Put	0.015981	0.015966	0.015193	0.113611
OTM Put	0.012260	0.012249	0.008034	0.083687

Direct-Low Gap

Scenario	Benchmark Direct-Low Gap	Hybrid Direct-Low Gap
ATM	-0.186%	+0.875%
ITM Put	-0.066%	+0.822%
OTM Put	-0.269%	+0.991%

BGK r=0, T=2 Comparison

BGK comparison across ATM, ITM, and OTM with $r = 0.0$ and $T = 2.0$.

Parameters

- $S_0 = 100$
- $r = 0.0$
- $T = 2.0$
- $\theta = 0.078$
- $\kappa_1 = 5.5$
- $\kappa_2 = 0.1$
- $v_0 = 0.114$
- $vp_0 = 0.110$
- $\delta_1 = 0.94$
- $\delta_2 = 0.94$
- $\xi_1 = 2.689$
- $\xi_2 = 0.502$
- $\rho_{12} = -0.982$
- $\rho_{13} = -0.727$
- $\rho_{23} = 0.59$
- Scenarios = ATM ($K = 100$), ITM put ($K = 110$), OTM put ($K = 90$)
- Exercise dates = 100
- Euler steps = 600
- Benchmark paths = 1,000,000
- Benchmark low paths = 1,000,000
- Hybrid paths = 20,000
- Hybrid low paths = 20,000
- Hybrid asset points = 181
- Hybrid asset low factor = 0.35

- Hybrid asset high factor = 3.00
- Hybrid volatility quantile = 0.997
- Hybrid FST pad factor = 4
- Hybrid FST batch size = 256
- Benchmark seeds = 2026, 2103
- Hybrid seeds = 2026, 2103

Scenario Summary

Direct

Scenario	K	Benchmark Direct	Hybrid Direct	Hybrid Direct Relative Error
ATM	100	15.537393	15.467526	0.450%
ITM Put	110	20.812594	20.796618	0.077%
OTM Put	90	11.200789	11.160237	0.362%

Low

Scenario	K	Benchmark Low	Hybrid Low	Hybrid Low Relative Error
ATM	100	15.506461	15.621656	0.743%
ITM Put	110	20.791793	20.980861	0.909%
OTM Put	90	11.170787	11.270917	0.896%

Metrics

Scenario	Benchmark Direct SE	Benchmark Low SE	Hybrid Direct SE	Hybrid Low SE
ATM	0.020775	0.020762	0.016108	0.143677
ITM Put	0.023205	0.023188	0.019759	0.164665
OTM Put	0.017856	0.017829	0.012533	0.120877

Direct-Low Gap

Scenario	Benchmark Direct-Low Gap	Hybrid Direct-Low Gap
ATM	-0.199%	+0.996%
ITM Put	-0.100%	+0.886%
OTM Put	-0.268%	+0.992%

BGK $r=0$, $T=1$, $\delta_1=\delta_2=0.5$ Comparison

BGK comparison across ATM, ITM, and OTM with $r = 0.0$, $T = 1.0$, and $\delta_1 = \delta_2 = 0.50$.

Parameters

- $S_0 = 100$
- $r = 0.0$
- $T = 1.0$
- $\theta = 0.078$
- $\kappa_1 = 5.5$
- $\kappa_2 = 0.1$
- $v_0 = 0.114$

- $vp0 = 0.110$
- $\delta1 = 0.50$
- $\delta2 = 0.50$
- $\xi1 = 2.689$
- $\xi2 = 0.502$
- $\rho12 = -0.982$
- $\rho13 = -0.727$
- $\rho23 = 0.59$
- Scenarios = ATM ($K = 100$), ITM put ($K = 110$), OTM put ($K = 90$)
- Exercise dates = 100
- Euler steps = 600
- Benchmark paths = 1,000,000
- Benchmark low paths = 1,000,000
- Hybrid paths = 20,000
- Hybrid low paths = 20,000
- Hybrid asset points = 181
- Hybrid asset low factor = 0.35
- Hybrid asset high factor = 3.00
- Hybrid volatility quantile = 0.997
- Hybrid FST pad factor = 4
- Hybrid FST batch size = 256
- Benchmark seeds = 2026, 2103
- Hybrid seeds = 2026, 2103

Scenario Summary

Direct

Scenario	K	Benchmark Direct	Hybrid Direct	Hybrid Direct Relative Error
ATM	100	10.548060	10.484748	0.600%
ITM Put	110	14.719728	15.035798	2.147%
OTM Put	90	7.512186	7.415154	1.292%

Low

Scenario	K	Benchmark Low	Hybrid Low	Hybrid Low Relative Error
ATM	100	10.519125	10.544421	0.240%
ITM Put	110	14.691320	14.798167	0.727%
OTM Put	90	7.496058	7.554493	0.780%

Metrics

Scenario	Benchmark Direct SE	Benchmark Low SE	Hybrid Direct SE	Hybrid Low SE
ATM	0.018176	0.018145	0.013958	0.125869
ITM Put	0.019787	0.019752	0.018057	0.144063
OTM Put	0.015707	0.015683	0.010424	0.105602

Direct-Low Gap

Scenario	Benchmark Direct-Low Gap	Hybrid Direct-Low Gap
ATM	-0.274%	+0.569%
ITM Put	-0.193%	-1.580%
OTM Put	-0.215%	+1.879%